TEXORA · COURSE NOTES · LECTURE SHEET

# Population Dynamics

*How populations grow, spread and change structure — the world's growth and its uneven map, the special case of Bangladesh, the urban flood and its consequences, and what population policy tries to do about all of it.*

HUM-2209 · SOCIOLOGY AND HUMAN ECOLOGY · TOPIC 6 · LEVEL 2 TERM 2

## 1. Basic ideas and measures

**Population dynamics** is the study of how a population's size, distribution and composition change over time. The changes come from only three sources: births, deaths and migration (Topic 4). The standard measures an examinee must handle:

- **Crude birth rate / death rate** — births (or deaths) per 1,000 people in a year.
- **Growth rate** — birth rate minus death rate (plus net migration), usually expressed as a percentage per year.
- **Fertility** — actual childbearing; the total fertility rate (TFR) is the average number of children a woman bears in her lifetime. About **2.1** is *replacement level* — the rate at which a generation just replaces itself.
- **Density** — persons per unit area; crude, but politically powerful.
- **Age–sex structure** — the shares of young, working-age and old, shown in the population pyramid (Figure 1).

The master pattern is the **demographic transition**: in stage 1, high birth and death rates roughly cancel and growth is slow. In stage 2, death rates fall (food, sanitation, medicine) while births stay high — population explodes. In stage 3, birth rates follow downward (education, urban life, contraception) and growth slows. In stage 4, both rates are low and the population is large but stable, with a growing share of elderly. Every country in this topic is somewhere on that staircase.

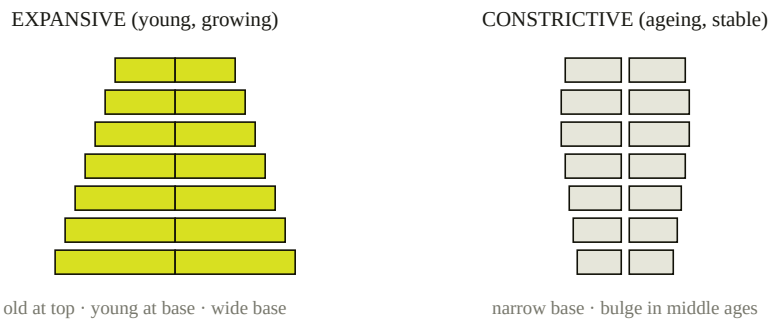



Figure 1 — Population pyramids (schematic): an expansive pyramid of a young, fast-growing population, and a constrictive one of an ageing, slow-growing population.

## 2. World population: growth and distribution

For most of human history the world's population grew almost imperceptibly — death rates matched birth rates and disasters repeatedly cut the total back. The curve bent upward with the agricultural and then the Industrial Revolutions, and bent sharply after 1950 as mortality fell across Asia, Africa and Latin America while fertility there was still high. The total passed one billion in the early nineteenth century, doubled and redoubled through the twentieth, and stands at **more than eight billion in the early 2020s**. Growth is now slowing as fertility falls almost everywhere, but the absolute numbers added each year remain large, and the increase is concentrated in the poorer countries least equipped to absorb it.

The distribution is strikingly uneven. **Asia holds well over half of humanity**; within it, the great river plains and deltas of South and East Asia support some of the densest rural populations on earth. Distribution follows habitability and livelihood: water, flat fertile land, moderate climate and industry concentrate people; deserts, high mountains, deep forest and polar lands repel them. Two consequences matter for this course: pressure on land is not a Bangladeshi eccentricity but the normal condition of the Asian plains; and the world's population is rapidly becoming an urban one — roughly half of humanity already lives in towns and cities, a share still rising (section 4).

## 3. Population dynamics in Bangladesh

Bangladesh is among the most densely populated large countries in the world: the 2022 census counted roughly **165 million people** (later estimates run somewhat higher) on a delta of about 148,000 square kilometres — **well over 1,000 persons per square kilometre**, and far more than that per unit of farmland once rivers and settlements are deducted. Density, not mere numbers, is the country's demographic signature, and its causes are those of section 2: a fertile, watered delta that can grow rice almost everywhere and reward every additional pair of hands — up to a point.

The dynamics have turned a historic corner. In the 1970s the average woman bore more than six children and the population was doubling in roughly a generation. Since then fertility has fallen steadily — to **about two children per woman, near replacement level** — one of the great social changes of the country's history, achieved through the family-planning programme (section 5), girls' schooling, women's employment including in the garment industry, and urbanisation. Yet numbers keep growing because of **population momentum**: the very large young generations of the high-fertility years are now the parents, so even replacement-level fertility adds millions for decades. Meanwhile the structure ages slowly, the working-age share is temporarily large (the much-discussed *demographic dividend* — an opportunity with an expiry date), and the movement from village to city continues without pause.

## 4. Urbanization: causes and consequences

**Urbanization** is the growth of the proportion of a population living in urban areas, driven mainly by rural-to-urban migration (the push and pull factors of Topic 4, now seen at national scale). In Bangladesh the pushes are land scarcity and landlessness, the fragmentation of holdings into plots too small to feed a family, river erosion, and the shortage of year-round rural work. The pulls are concentrated where industry is: garment and other factory employment around Dhaka and Chattogram, construction, transport, domestic and service work, plus education, hospitals and the simple reputation of the capital as the place where money circulates.

| POSITIVE CONSEQUENCES | NEGATIVE CONSEQUENCES |
|---|---|
| Labour reaches industry; national output and exports grow | Housing shortage answered by sprawling slums and informal settlements |
| Remittances and wages flow back, transforming villages | Water, sanitation, drainage and transport overwhelmed; congestion and flooding worsen |
| Services — health, education, markets — become accessible and cheap to deliver per head | Air, water and noise pollution concentrate (Topic 5); health hazards multiply |
| Fertility falls; women's employment and independence rise | New urban poverty, insecure informal work, crime and social strain |
| Ideas, skills and ambitions circulate faster | Prime farmland at the city's edge is built over, cutting the food base |

The policy problem is not how to stop urbanization — no country has — but how to spread it (secondary cities, district towns) and service it (housing, water, transport) before the informal city hardens into permanence.

## 5. Population policy in Bangladesh

A **population policy** is the set of measures by which a government tries to influence the size, growth, distribution or composition of its population. Bangladesh adopted an explicit population programme early — family planning services began in the 1960s–70s and were made a national priority after independence, when population was officially named the country's "number one problem." Its machinery: a field-worker network taking contraception and maternal-child health door to door, clinics and satellite services, mass-media motivation (the long-running message that a small family is a happy family — "two children are enough"), incentives and later a rights-based emphasis on informed choice.

Its present aims go beyond slowing growth:

- Hold fertility at about replacement level and meet the remaining unmet need for family planning.
- Reduce maternal and child mortality — population policy merged with health policy.
- Delay early marriage and first childbirth, which remain common and raise both fertility and health risks.
- Manage distribution: relieve Dhaka's extreme primacy by developing other urban centres and keeping rural areas viable.
- Prepare for ageing: as the structure shifts, pensions, chronic disease and old-age support move onto the agenda.
- Use the demographic dividend: educate and employ the large working-age generation while it lasts.

The judged result, in the standard assessment: the fertility transition was a genuine achievement against difficult odds; the unfinished business is urban management, adolescent fertility, and the coming age-structure turn.

## 6. Glossary — key terms of this topic

| TERM | MEANING |
|---|---|
| Total fertility rate (TFR) | The average number of children a woman would bear over her lifetime at current rates; about 2.1 is replacement level. |
| Demographic transition | The passage from high birth and death rates to low ones, through a phase of rapid growth when deaths fall first. |
| Population momentum | Continued growth after fertility reaches replacement level, because large young generations become the parents. |
| Demographic dividend | The economic opportunity of a period when working-age people greatly outnumber dependants. |

| | |
|---|---|
| Population pyramid | A bar diagram of a population's age and sex structure; its shape reveals growth, stability or decline. |
| Urbanization | The rise in the share of a population living in urban areas. |

## 7. Key points and summary

- Populations change only through births, deaths and migration; the demographic transition organises these into a universal sequence ending, so far, in low growth and ageing.
- The world passed eight billion in the early 2020s; growth is slowing but concentrated in poorer countries, and distribution follows water, land and livelihood — Asia holds the majority.
- Bangladesh: roughly 165 million (2022 census) at over 1,000 per km²; fertility fallen to about replacement; momentum and a young structure still add numbers; the dividend window is open but temporary.
- Urbanization is pulled by industry and pushed by land hunger; its benefits (output, income, services) and costs (slums, strain, pollution) arrive together.
- Bangladesh's population policy, one of the world's most determined, cut fertility dramatically; its frontier is now cities, adolescents and ageing.

## 8. Practice

Answer in your own words; brief model answers follow.

1. A country has a birth rate of 18 and a death rate of 7 per 1,000, with no migration. Find its growth rate as a percentage, and state its likely transition stage with a reason.
2. Explain population momentum: why does Bangladesh keep adding millions each year although fertility is about replacement level?
3. Look at Figure 1. Which pyramid belongs to a country with many school places to build, and which to a country that must plan pensions? Justify from the shapes.
4. “Density, not numbers, is Bangladesh's demographic signature.” Explain in four sentences.
5. A village family loses its plot to river erosion; the son finds work in a Chattogram factory. Classify the forces at work and name two consequences — one at each end of his journey.
6. State two successes claimed for Bangladesh's population policy and two tasks it still faces.

### Answer key

1. Natural increase = 18 − 7 = 11 per 1,000 = 1.1% per year. With deaths already low and births moderately low and falling, it fits late stage 3 moving into stage 4 of the transition.

2. Past high fertility produced very large young cohorts. As they reach childbearing age, the sheer number of parents is so great that even two children each yields more births than the deaths in the smaller older generations — growth continues for decades after replacement fertility is reached.
3. The expansive pyramid (wide base) means large child cohorts — school places, then jobs, must be built in waves. The constrictive pyramid (narrow base, heavy middle and top) means few young workers supporting many elderly — pensions and elder care dominate planning.
4. Many countries have larger populations; few crowd them onto so small and river-cut a land. Over 1,000 persons per km² overall — and far more per unit of farmland — means every policy (food, housing, flood control, jobs) is decided under land pressure that most countries never face.
5. Erosion is a push factor; factory wages a pull factor. At the origin: a family loses its livelihood and a village loses a worker (but may gain remittances). At the destination: industry gains labour while housing and services in the factory district strain further.
6. Successes: fertility brought from over six children per woman to about replacement level; a nationwide family-planning and maternal-health service built and sustained. Remaining tasks: managing explosive urban growth and Dhaka's primacy; reducing early marriage and adolescent childbearing (accept also: preparing for ageing).

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